

COMS 4720/5720: Homework 3
Due: February 20th, 11:59pm
Total Points: 40

Late submission policy. An assignment that is submitted one day late will get a penalty of 10%. An assignment that is submitted two days late will get a penalty of 20%. Assignments submitted after two days will not be graded and will get no points. For example, if an assignment is due on Friday by midnight, a submission on Saturday will be penalized by 10%, and a submission on Sunday will be penalized by 20%. A submission on Monday will get no points.

Submission format. Homework solutions must be **handwritten**. You may either write directly on a tablet or write on paper and scan your work. In both cases, the final submission must be compiled into a **single PDF file**. Typed solutions are not permitted unless you obtain prior, explicit permission from the instructor. We reserve the right **NOT** to grade homework that does not follow these requirements.

The submission file must be named as follows:

- <Your-net-id>-4720-hw3.pdf for COMS 4720
- <Your-net-id>-5720-hw3.pdf for COMS 5720.

For instance, if your netid is **asterix** and you are enrolled COMS 4720, then your submission file should be named **asterix-4720-hw3.pdf**. Each student must hand in their own assignment. If you discussed the homework or solutions with others, a list of collaborators must be included with each submission. Each of the collaborators has to write the solutions in their own words (copies are not allowed). Also, recall the AI policy:

- You may **not use AI tools to generate complete solutions** to assignments, coding problems, or any assessed work that is intended to measure your independent knowledge and skills.
- Whenever you use AI tools to assist in your assignment, you must **include a paragraph at the end of the assignment explaining what you used the AI for and include all the prompts you used to get the results**. Failure to do so is in violation of the academic honesty policies.

1. **(15 pts)** The heuristic path algorithm is a best-first search algorithm in which the evaluation function is $f(n) = (2 - w)g(n) + wh(n)$, where $g(n)$ is the path cost from the start node to n and $h(n)$ is a heuristic estimate of the cost from n to a goal.

- (a) **(5 pts)** State the necessary condition on the values of the parameter w under which the heuristic path algorithm is guaranteed to be complete.

Best-first search will be complete if its cost function is positive. The cost function $f(n)$ can go negative when $(2 - w) < 0$ or $w < 0$. So, to guarantee completeness, $0 \leq w \leq 2$ is necessary.

- (b) **(5 pts)** State the necessary condition on the values of the parameter w under which the heuristic path algorithm is guaranteed to return an optimal solution, assuming that h is admissible?

Best-first search will be optimal if its cost function is positive and the heuristic function is admissible. The heuristic function is admissible when $0 \leq w \leq 1$ and the cost function $f(n)$ will keep positive when $0 \leq w \leq 2$. Thus, when $0 \leq w \leq 1$, the best-first search is optimal.

- (c) **(5 pts)** What kind of search does this perform for $w = 0$, $w = 1$, and $w = 2$?

$w = 0$: Uniform-cost search.

$w = 1$: A* search.

$w = 2$: Greedy search (best-cost search).

2. **(15 pts)** There are n vehicles which occupy squares $(1, 1)$ through $(n, 1)$ (i.e., the leftmost column) of an $n \times n$ grid. The vehicles must be moved to the rightmost column but in reverse order; so the vehicle i that starts in $(i, 1)$ must end up in $(n - i + 1, n)$. On any time step, every one of the n vehicles can move one square up, down, left, or right, or stay put; but if a vehicle stays put, one other adjacent vehicle (but not more than one) can hop over it. No two vehicles can occupy the same square.

- (a) **(2.5 pts)** What is the size of the state space as a function of n ?

A state is a labeled placement of n distinct vehicles on n^2 squares (one per square).
Hence

$$|S| = P(n^2, n) = \frac{n^2!}{(n^2 - n)!} = (n^2)(n^2 - 1) \cdots (n^2 - n + 1) \in \Theta((n^2)^n).$$

- (b) **(2.5 pts)** What is the branching factor as a function of n ?

Per time step, each vehicle may move up, down, left, right, or stay (subject to boundaries, occupancy, and at most one hop enabled by a staying neighbor). This gives the bound

$$b \leq 5^n$$

- (c) **(5 pts)** Suppose that vehicle i is at (x_i, y_i) ; write a nontrivial admissible heuristic h_i for the number of moves required to get the vehicle to its goal location $(n - i + 1, n)$, assuming no other vehicles are on the grid.

For vehicle i at (x_i, y_i) with goal $(n - i + 1, n)$, the Manhattan distance

$$h_i = |(n - i + 1) - x_i| + |n - y_i|$$

is admissible for the single-vehicle problem (empty grid): each move changes the L_1 distance by at most 1, so h_i never overestimates the steps needed.

- (d) **(5 pts)** Which of the following heuristics are admissible for the problem of moving all n vehicles to their destinations? Explain.

$$\begin{aligned} H_{\Sigma} &= \sum_{i=1}^n h_i \\ H_{\max} &= \max_i h_i \\ H_{\min} &= \min_i h_i \end{aligned}$$

Only $H_{\min}$ is admissible for the joint problem under the given move/hop rules:

- $H_{\min}$ is a valid lower bound on the number of time steps: at least one vehicle must require $\min_i h_i$ moves, and parallel progress (including hops) cannot reduce the makespan below this minimum.
- $H_{\max}$ can overestimate: there are layouts where the farthest vehicle reaches its goal faster than $H_{\max}$ via hops enabled by neighbors that stay, so $H_{\max} >$ optimal cost in those cases.
- H_{Σ} can also overestimate (and always dominates $H_{\max}$), so it is not admissible.

3. **(5 pts)** Prove or disprove: If a heuristic is consistent, it must be admissible.

We will prove the claim.

A heuristic is consistent iff, for every node n and every successor n' of n generated by any action a ,

$$h(n) \leq c(n, a, n') + h(n')$$

One simple proof is by induction on the number k of nodes on the shortest path to any goal from n . For $k = 1$, let n' be the goal node; then $h(n) \leq c(n, a, n')$. For the inductive case, assume n' is on the shortest path k steps from the goal and that $h(n')$ is admissible by hypothesis; then

$$h(n) \leq c(n, a, n') + h(n') \leq c(n, a, n') + h^*(n') = h^*(n)$$

so $h(n)$ at $k + 1$ steps from the goal is also admissible.

4. **(5 pts)** Prove or disprove: If a heuristic is admissible, it must be consistent.

We will disprove the claim by constructing a heuristic that is admissible but not consistent.

Consider a search problem with a start state s , an intermediate state a , and a goal state g . The action costs are $c(s, a) = 1$, $c(a, g) = 1$, and $c(s, g) = 3$.

In this case, the true optimal costs to the goal are $h^*(s) = 2$, $h^*(a) = 1$, and $h^*(g) = 0$.

Consider a heuristic h with $h(s) = 2$, $h(a) = 0$, and $h(g) = 0$. It is clear that $h(n) \leq h^*(n)$ for all n so h is admissible. However, $h(s) = 2$ but $c(s, a) + h(a) = 1$. So $h(s) > c(s, a) + h(a)$ and by definition h is not consistent.